

Homework Project 3: AI/ML Part 1

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1 EXPLORING THE DATA

1.1 Dataset Cleanup

- I checked for null values in row and columns, most columns only have 1 or 2 null values which are safe to ignore
- Found out the row with toxicity 52671 is null completely. so I removed that row, and then assigned 52671 as wiki_id for the next row as it was empty. This could have been a misplacement of wiki_id.

1.2 Identifying protected classes and subgroups

Protected Class	# Subgroups	Members
Sexual Orientation	9	lesbian, gay, bisexual, queer, lgbt, lgbtq, homosexual, straight, heterosexual
Gender	5	male, female, nonbinary, transgender, trans
Race	8	black, white, asian, african, african american, indian, middle eastern
Ethnicity	5	hispanic, latino, latina, latinx, european
National Origin	5	mexican, canadian, american, chinese, japanese
Religion	8	christian, muslim, jewish, buddhist, catholic, protestant, sikh, taoist
Age	8	old, older, young, younger, teenage, millennial, middle aged, elderly
Disability	3	blind, deaf, paralyzed

2 CALCULATING TOXICITY CORRELATIONS

2.1 Creating reduced data set

The original dataset had 76564 rows, we removed 864 rows that had all FALSE values for every column, and we ended up with 75700 rows in the reduced dataset.

2.2 Identifying ordering schema for each protected class

1. Gender / Sex

Value	Subgroup
0	FALSE
1	female
2	male
3	nonbinary
4	trans
5	transgender

2. Sexual Orientation

Value	Subgroup
0	FALSE
1	bisexual
2	gay
3	heterosexual
4	homosexual
5	lesbian
6	lgbt
7	lgbtq
8	queer
9	straight

3. Race

Value	Subgroup
0	FALSE
1	african
2	african american
3	asian
4	black
5	indian
6	middle eastern
7	white

4. Ethnicity

Value	Subgroup
0	FALSE
1	european
2	hispanic
3	latina
4	latino
5	latinx

5. National Origin

Value	Subgroup
0	FALSE
1	american
2	canadian
3	chinese
4	japanese
5	mexican

6. Religion

Value	Subgroup
0	FALSE
1	buddhist
2	catholic
3	christian
4	jewish
5	muslim
6	protestant
7	sikh
8	taoist

7. Age

Value	Subgroup
0	FALSE
1	elderly
2	middle aged
3	millennial
4	old
5	older
6	teenage
7	young
8	younger

8. Disability

Value	Subgroup
0	FALSE
1	blind
2	deaf
3	paralyzed

I converted each subgroup's false value to zero and the subgroups were numbered alphabetically, in alphabetical order according to the column name. I choose to use alphabetical order to avoid bias for each sub group. These numbers are really just for labeling. So for limitation, is that even though these numbers are just for labeling purposes, the correlation might still treat them as if they have meaningful order based on the number. For example, it might think that the value 1 has more importance than the value 2 or 3.

2.4 Calculating the correlation between the protected class columns and TOXICITY

Correlation with TOXICITY		
Protected Class	Correlation with TOXICITY	Strength
Gender	0.051	very weak
Sexual Orientation	0.131	very weak
Race	-0.083	very weak
Ethnicity	0.002	very weak
National Origin	-0.085	very weak
Religion	0.018	very weak
Age	-0.033	very weak
Disability	0.012	very weak

2.5 Subgroup values vs Toxicity

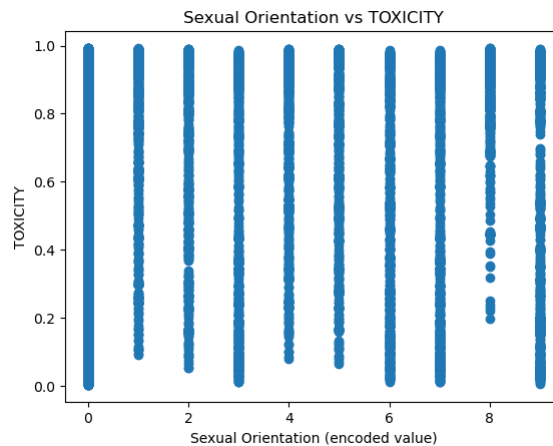


Figure 1—Sexual Orientation and Toxicity

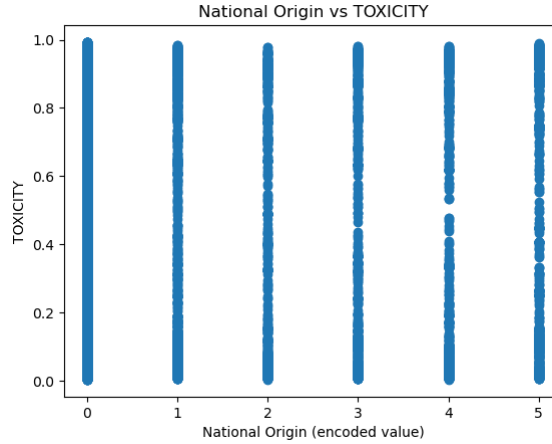


Figure 2—National Origin and Toxicity

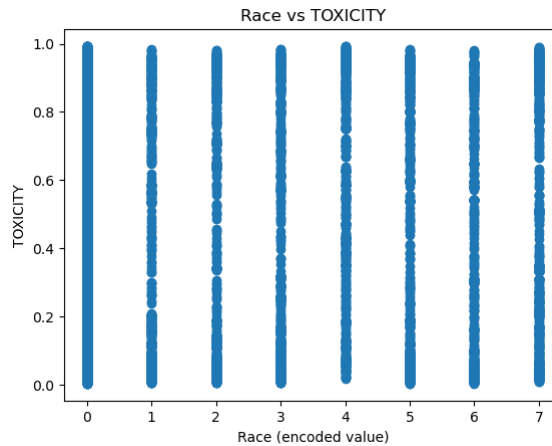


Figure 3—Race and Toxicity

2.6 Do the observed correlations describe the relationship between comments that mention a subgroup and their toxicity?

The observed correlations do not really describe the relationship between comments mentioning a subgroup and their toxicity. Although Sexual Orientation, National Origin, and Race had the highest absolute values, all were very weak (below 0.20). The scatter plots show the same pattern for all three: at each encoded subgroup value, toxicity is distributed across nearly the full 0–1 range with no clear linear trend. Therefore, correlation alone is a poor measure in this context, using mean toxicity by individual subgroup will provide a clearer picture of bias in the dataset in the later section.

3 ANALYZING TOXICITY ON THE REDUCED DATASET

3.1 Population mean, population standard deviation, and range of values around the mean that includes 95% TOXICITY

- Mean: 0.550101802360502
- Standard deviation: 0.36185007482613896
- Range: -0.17359834729177592 to 1.27380195201278

3.2 Mean, standard deviation, and margin of error of 10% dataset

- Mean: 0.556322766956539
- Standard deviation: 0.3621200635972302
- Margin of error: 0.008324049303570698

3.3 Mean, standard deviation, and margin of error of 60% dataset

- Mean: 0.5513382534439233
- Standard deviation: 0.3616727177771927
- Margin of error: 0.00832404930357069

4 ANALYZING TOXICITY FOR A CHOSEN PROTECTED CLASS

4.1 Mean, standard deviation of TOXICITY associated with the protected class

- Mean: 0.5984416701509908
- Standard deviation: 0.34264019191400086

4.2 Mean, standard deviation, margin of error, and explanation of 10% dataset

- Mean: mean: 0.6169930197246558
- Standard deviation: 0.34520410706448335
- Margin of error: 0.024424886827611163
- Did protected class sample mean lie within the margin of error: Yes
- explanation: This sample mean lies within the margin of error so it shows that the 10% gender subsample is consistent with the full gender population mean and the differences are within expected sampling variation.

4.3 Mean, standard deviation, margin of error, and explanation of 60% dataset

- Mean: 0.5986179938276786
- Standard deviation: 0.34419413477263155

- Margin of error: 0.010284766926133681
- Did protected class sample mean lie within the margin of error: Yes
- explanation: This 60% gender sample mean also lies within the margin of error around the population mean but it has lower mean and lower margin of error compared to 10% dataset sample. This shows that larger sample size reduces sampling error, so the sample mean is very close to the true gender population mean.

5 ANALYZING TOXICITY FOR THE SUBGROUPS OF THE CHOSEN PROTECTED CLASS

5.1 Mean and standard deviation of TOXICITY associated with each subgroup that is a member of the protected class selected in Step 4

Gender Subgroup Statistics

Subgroup	Mean	Std Dev
female	0.603425	0.336896
male	0.585090	0.337665
nonbinary	0.555164	0.344433
trans	0.507426	0.379844
transgender	0.741103	0.256646

5.2 Mean, standard deviation, margin of error, and explanation of each subgroup of the protected class of 10% dataset

Sample Statistics and Margin of Error

Subgroup	Sample Mean	Sample Std Dev	Margin of Error	Within MOE
female	0.634241	0.335264	0.053685	Yes
male	0.544272	0.358891	0.058803	Yes
nonbinary	0.578987	0.345887	0.055745	Yes
trans	0.528199	0.385652	0.061953	Yes
transgender	0.767050	0.238982	0.035141	Yes

Explanation: The 10% sample mean falls within the margin of error built from Step 5.1 population mean and standard deviation. For example, female's sample

mean (0.634) is higher than its population mean (0.603), and male's sample mean (0.544) is lower than its population mean (0.585), but both differences are smaller than the comparison margin of error (0.05). The margin of error is wider for smaller samples because the 10% sample is relatively small, so sample means can drift a bit from the true population mean

5.3 Mean, standard deviation, margin of error, and explanation of each subgroup of the protected class of 60% dataset

Sample Statistics and Margin of Error

Subgroup	Sample Mean	Sample Std Dev	Margin of Error	Within MOE
female	0.606678	0.335668	0.022378	Yes
male	0.575580	0.341450	0.023086	Yes
nonbinary	0.558760	0.345541	0.023191	Yes
trans	0.513909	0.383310	0.025413	Yes
transgender	0.736858	0.261024	0.017334	Yes

Explanation: In the 60% sample, every subgroup's sample mean is much closer to its Step 5.1 population mean than in the 10% sample. For example, transgender's sample mean (0.737) is almost identical to its population mean (0.741), with a difference of only about 0.004. This is expected because the 60% sample is much larger, so the sample mean is a more reliable estimate of the population mean.

6 PLOTS AND TOXICITY ANALYSIS

6.1 Graph containing mean and standard deviation from Step 3, 4, and 5

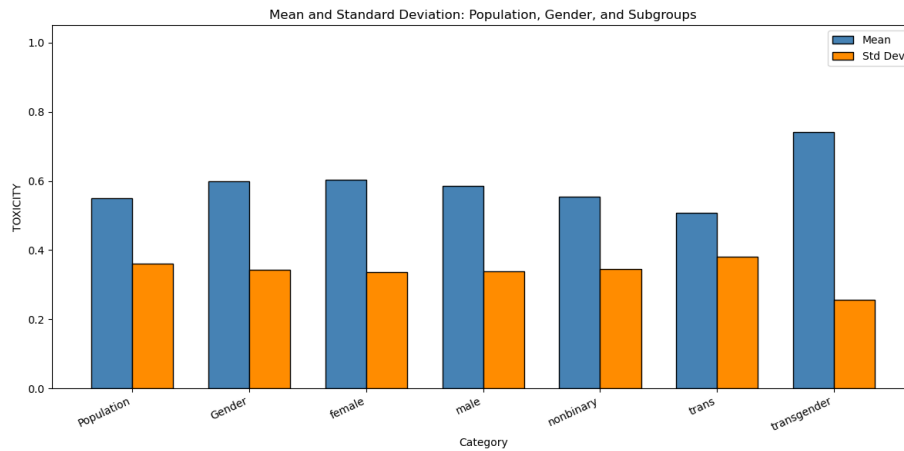


Figure 4—Mean and Standard deviation of all categories

6.2 Which subgroup has the highest TOXICITY value? Why?

The "transgender" subgroup has the highest TOXICITY value. It might be because the sources used to form the dataset mostly have toxic comments using the words "transgender". This doesn't mean that having the word "transgender" in the comments automatically makes the comments toxic, it just happens to be in this case due to the sources being used.

6.3 Which subgroup has the lowest TOXICITY value? Why?

The "trans" subgroup has the lowest TOXICITY value. It might be because the sources used to form the dataset mostly have neutral or non-toxic comments using the words "trans". Again, this doesn't mean that having the word "trans" in the comments automatically makes the comments neutral or non-toxic, it just happens to be in this case due to the sources being used. A comment can also be toxic while having the word "trans".

6.4 Which subgroup has the largest difference in TOXICITY value when compared to the population mean? Why?

The transgender subgroup has the largest difference. This gap is likely because "transgender" is a more specific term than others like "male" or "female." It

could be that the comments that use it may more often appear in debates or disputes, which tend to get higher toxicity scores.

6.5 What type of human or systemic bias may be present in this dataset?

The bias present is gender bias since comments that mention the identity term “transgender” receive much higher average toxicity scores than the overall population and than comments mentioning “trans,” even though both terms refer to related gender-identity topics.

6.5.1 *How can this bias be quantified?*

We quantified this using subgroup mean toxicity, differences from the population mean, and comparisons across gender subgroups as shown in Steps 4 to 6.

6.5.2 *How might it be reduced in future datasets or models?*

To reduce this bias, future datasets should include more comments from diverse sources, especially neutral comments that mention gender identity terms like “transgender.” If the data mostly contains those words in already-toxic comments, the model will learn the wrong association.